WORKED SOLUTION

STEP III 2008, Question 4

Pure • Inverse Hyperbolic Inequalities

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q4

The key idea

Put $x = \cosh y$. The half-angle formula turns $\tanh(y/2)$ into an algebraic function of x, so a trivial inequality about tanh becomes a non-trivial one about arcosh. After that the question runs a little machine: integrate an inequality from 1 to x, rearrange, and get a sharper inequality. Running the machine twice gives (ii) and then (iii).

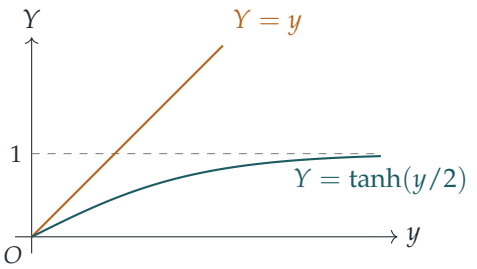
Solution

Throughout, write $A(x) = \operatorname{arcosh} x$ and $S(x) = \sqrt{x^2 - 1}$ for $x \geq 1$. Two derivatives we shall need are

$$\frac{\mathrm{d}}{\mathrm{d}x} \operatorname{arcosh} x = \frac{1}{\sqrt{x^2 - 1}}, \qquad \frac{\mathrm{d}}{\mathrm{d}x} \sqrt{x^2 - 1} = \frac{x}{\sqrt{x^2 - 1}} \qquad (x > 1).$$

Part (i)

The curve $Y = \tanh(y/2)$ passes through the origin with gradient $\frac{1}{2}$, is concave for $y > 0$ and never reaches 1, whereas the line $Y = y$ has gradient 1:



The line lies above the curve for every $y > 0$. To make this precise, let $g(t) = t - \tanh t$. Then $g(0) = 0$ and

$$g'(t) = 1 - \operatorname{sech}^2 t = \tanh^2 t > 0 \quad \text{for } t > 0,$$

so g is strictly increasing and $\tanh t < t$ for $t > 0$. With $t = y/2$,

$$\tanh(y/2) < \tfrac{1}{2}y < y \qquad (y > 0).$$

Now let $x > 1$ and put $y = \operatorname{arcosh} x$, so $y > 0$, $\cosh y = x$ and $\sinh y = +\sqrt{x^2 - 1}$ (positive because $y > 0$). Using $\sinh y = 2\sinh \frac{y}{2}\cosh \frac{y}{2}$ and $1 + \cosh y = 2\cosh^2 \frac{y}{2}$,

$$\tanh \frac{y}{2} = \frac{\sinh y}{1 + \cosh y} = \frac{\sqrt{x^2 - 1}}{x + 1} = \frac{x^2 - 1}{(x+1)\sqrt{x^2 - 1}} = \frac{x - 1}{\sqrt{x^2 - 1}}.$$

So $y > \tanh(y/2)$ becomes

$$\operatorname{arcosh} x > \frac{x - 1}{\sqrt{x^2 - 1}} \qquad (x > 1),$$

which is $(*)$.

Part (ii)

Fix $x > 1$ and integrate $(*)$, with t as the variable, over $[1, x]$.

Ask yourself The right-hand side of $(*)$ has $\sqrt{t^2 - 1}$ in the denominator. Is it safe to integrate from $t = 1$, and does integrating keep the inequality *strict*?

Yes on both counts. For $t > 1$, $\dfrac{t-1}{\sqrt{t^2-1}} = \sqrt{\dfrac{t-1}{t+1}}$, which tends to 0 as $t \to 1$, so both sides are continuous on $[1, x]$. Their difference is continuous, and strictly positive on $(1, x]$, so its integral over $[1, x]$ is strictly positive.

The left-hand side, integrating by parts with $u = \operatorname{arcosh} t$:

$$\int_1^x \operatorname{arcosh} t \, \mathrm{d}t = \Big[t \operatorname{arcosh} t\Big]_1^x - \int_1^x \frac{t}{\sqrt{t^2 - 1}} \, \mathrm{d}t = x\,A(x) - S(x),$$

since $\operatorname{arcosh} 1 = 0$ and $\sqrt{1^2 - 1} = 0$.

The right-hand side splits into two standard pieces:

$$\int_1^x \frac{t-1}{\sqrt{t^2-1}} \, \mathrm{d}t = \int_1^x \frac{t}{\sqrt{t^2-1}} \, \mathrm{d}t - \int_1^x \frac{1}{\sqrt{t^2-1}} \, \mathrm{d}t = S(x) - A(x).$$

Therefore

$$x\,A(x) - S(x) > S(x) - A(x) \quad \Longrightarrow \quad (x+1)A(x) > 2S(x).$$

Dividing by $x + 1 > 0$ and simplifying as in part (i),

$$\operatorname{arcosh} x > \frac{2\sqrt{x^2 - 1}}{x + 1} = 2\,\frac{x - 1}{\sqrt{x^2 - 1}} \qquad (x > 1),$$

as required. This is exactly twice the bound in $(*)$.

Part (iii)

Part (ii) says $\operatorname{arcosh} t > 2 \cdot \dfrac{t-1}{\sqrt{t^2-1}}$ for $t > 1$: the same right-hand side as $(*)$, doubled. So we run the same machine again. Integrating over $[1, x]$, with the same justification as before, and using the two integrals already found,

$$x\,A(x) - S(x) > 2\big(S(x) - A(x)\big) \quad \Longrightarrow \quad (x+2)A(x) > 3S(x).$$

Since $x + 2 > 0$,

$$\operatorname{arcosh} x > 3\,\frac{\sqrt{x^2-1}}{x+2} \qquad (x > 1),$$

which is the required result.

Ask yourself Why stop here? Can we integrate once more?

The machine needs the right-hand side to be a multiple of $\dfrac{t-1}{\sqrt{t^2-1}}$, whose integral we know. The new bound $3\sqrt{t^2-1}/(t+2)$ isn't of that form, so the trick runs out.

As an independent check (an "otherwise" route), let $f(x) = A(x) - 3S(x)/(x+2)$. Then $f(1) = 0$ and, by the quotient rule,

$$f'(x) = \frac{(x+2)^2 - 3x(x+2) + 3(x^2-1)}{(x+2)^2\sqrt{x^2-1}} = \frac{(x-1)^2}{(x+2)^2\sqrt{x^2-1}} > 0 \quad (x > 1),$$

so f is strictly increasing and $f(x) > 0$ for $x > 1$, as before.

Checks

At $x = 2$: $\operatorname{arcosh} 2 = \ln(2+\sqrt{3}) \approx 1.3170$, while the three lower bounds are $1/\sqrt{3} \approx 0.577$, $2/\sqrt{3} \approx 1.155$ and $3\sqrt{3}/4 \approx 1.299$. Each bound beats the last and all lie below 1.3170. Near $x = 1$ the bounds become very good: at $x = 1.001$, $\operatorname{arcosh} x \approx 0.04471763$, the bound in (ii) is short by about 7×10^{-6} and the bound in (iii) by only about 10^{-9}, because each matches $\operatorname{arcosh} x$ to higher order in $x - 1$. A numerical check of the two integrals (by midpoint quadrature at $x = 1.5$, 2, 5) and of all three inequalities at 10 000 random $x > 1$ found no discrepancy.

Common slips

- Taking $\sinh(\operatorname{arcosh} x) = \pm\sqrt{x^2-1}$ without fixing the sign. It is the positive root because $\operatorname{arcosh} x > 0$.
- Integrating from 0 instead of 1: arcosh is only defined for $x \geq 1$, and the lower limit 1 is what makes the boundary terms vanish.
- Forgetting to say why the integrated inequality is still strict.

Beyond STEP

The bounds get better because each integration makes the error vanish to one higher order at $x = 1$; this is the start of a sequence of increasingly accurate rational-type approximations, close cousins of Padé approximants. The principle used throughout (if $f(a) = 0$ and $f' > 0$, or the integrated form of that, then $f > 0$) is the mean value theorem in disguise, and it underlies a great deal of analysis. Abbott's *Understanding Analysis* treats the mean value theorem and integration rigorously.

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